

TISSUES

Class 9 | Science | Chapter 6

WHAT IS A TISSUE?

A tissue is a group of cells similar in structure that work together to perform a specific function.

In multicellular organisms, cells show 'division of labour' -- groups of cells specialise into tissues for particular jobs.

PLANT TISSUES -- OVERVIEW

Plant tissues are of two main types: Meristematic (dividing) and Permanent (non-dividing, differentiated).

Meristematic Tissue

Cells that actively divide to produce new cells, found at growing regions (root tip, shoot tip). Types: Apical, Lateral, Intercalary.

Permanent Tissue

Cells that have lost the ability to divide; formed from meristematic tissue that has become specialised (differentiated).

PLANT TISSUES

Simple & Complex Permanent Tissues

SIMPLE PERMANENT TISSUE

Tissue	Feature / Function
Parenchyma	Thin-walled, loosely packed; storage of food; basic packing tissue
Collenchyma	Thickened at corners; gives flexibility & mechanical support to young stems
Sclerenchyma	Thick, lignified, dead cells; gives strength & rigidity (e.g. in husk of coconut, seeds)
Epidermis	Outermost protective layer of the plant; often covered with cuticle to prevent water loss

COMPLEX PERMANENT TISSUE

Tissue	Function
Xylem	Conducts water & minerals upward from roots; provides mechanical support (tracheids, vessels, xylem parenchyma, xylem fibres)
Phloem	Transports food (prepared by leaves) to all parts of the plant (sieve tubes, companion cells, phloem parenchyma, phloem fibres)

Xylem & phloem together are called 'vascular tissue' and form vascular bundles.

ANIMAL TISSUES

Epithelial, Connective, Muscular & Nervous

EPITHELIAL TISSUE

Covers/lines body surfaces and organs; forms a protective barrier; cells tightly packed with little intercellular space.

TYPES OF EPITHELIAL TISSUE

SQUAMOUS



COLUMNAR



CUBOIDAL



Squamous -- lining of blood vessels/mouth. Columnar -- lining of intestine (absorption). Cuboidal -- lining of kidney tubules.

CONNECTIVE TISSUE

Type	Example / Function
Blood	Fluid matrix (plasma); transports gases, nutrients, wastes
Bone	Hard matrix; gives shape & support to the body
Cartilage	Smooth, flexible matrix; found in nose, ear, joints
Ligament	Connects bone to bone; elastic
Tendon	Connects muscle to bone; strong, fibrous, non-elastic
Areolar	Fills space inside organs, supports internal organs

MUSCULAR TISSUE

Type	Feature
Striated (skeletal)	Voluntary; attached to bones; long, cylindrical, unbranched
Unstriated (smooth)	Involuntary; found in stomach, intestine; spindle-shaped
Cardiac	Involuntary; found only in heart; branched, works rhythmically

NERVOUS TISSUE

Made up of neurons; receives stimuli and transmits impulses across the body -- found in brain, spinal cord, nerves.

SUMMARY & REVISION

Key Terms + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Meristematic tissue** -> actively dividing plant tissue
- **Permanent tissue** -> non-dividing, differentiated plant tissue
- **Xylem** -> conducts water & minerals (upward)
- **Phloem** -> conducts food (both directions)
- **Epithelial tissue** -> covers/lines surfaces
- **Connective tissue** -> connects & supports body parts
- **Muscular tissue** -> enables movement
- **Nervous tissue** -> carries nerve impulses

PLANT VS ANIMAL TISSUE

Feature	Plant Tissue	Animal Tissue
Growth	Localised (at meristems)	Uniform, throughout the body
Main role	Support, storage, transport	Movement, repair, control

ONE-LINE RECAP FOR REVISION

Plant tissues: Meristematic (dividing) + Permanent (simple & complex).

Xylem carries water up; Phloem carries food both ways.

Animal tissues: Epithelial, Connective, Muscular, Nervous.

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